

Yiming Zhang
Curriculum Vitae

Institute for Rock Magnetism
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ACADEMIC APPOINTMENT

Postdoctoral Associate, Institute for Rock Magnetism, University of Minnesota, Twin Cities Sep 2024-

Postdoctoral Scholar, Earth and Planetary Science, University of California, Berkeley May-Sep 2024

EDUCATION

Ph.D., Earth and Planetary Science, University of California, Berkeley 2024

B.A., cum laude., Geology, Occidental College 2019

PUBLICATIONS
(* indicates mentored student)

Zhang, Y., Swanson-Hysell, N.L., Domeier, M., Gallo, L., Sapienza, F., Osorio Afanador, D.A, (2026), Enriching paleogeographic reconstruction through the integration of Bayesian age models into site-based apparent polar wander paths. *JGR Solid Earth*.

Zhang, Y., Hodgin, E.B., Schmitz, M.D., Schwartz, D., Mohr, M.T., Crowley, J.L, Swanson-Hysell, N.L., (2026), Protracted high-grade metamorphism and slow cooling in the Adirondacks recalibrate Neoproterozoic global paleogeography. *Tectonics*. <https://doi.org/10.1029/2025TC009333>

Kahn, L.X.*, **Zhang, Y.**, Finnegan, Seth., Hodgin, E.B., Swanson-Hysell, N.L., (2025), The stratigraphic record of the arrival of the Sacramento and San Joaquin Rivers to the California coast. *Journal of Sedimentary Research*. <https://doi.org/10.2110/jsr.2024.153>

Fuentes, A.J., Courtney-Davies, Liam., Flowers, R.M., **Zhang, Y.**, and Swanson-Hysell, N.L., (2025), Evolution of iron formation to ore during Ediacaran to early Paleozoic tectonic stability. *Earth and Planetary Science Letters*. <https://doi.org/10.1016/j.epsl.2025.119621>

Swanson-Hysell, N.L., **Zhang, Y.**, Macdonald, F.A., Koran, I., Tasistro-Hart, A., and Jay, A., (2025), Oman was on the northern margin of a wide late Tonian Mozambique Ocean. *Geology*. <https://doi.org/10.1130/G53450.1>

Wenk, HR., Kattamalavadi, A., **Zhang, Y.**, Kennedy, E.R., Borkiewicz, O., (2025), Exploring microstructures and anisotropies of serpentinites. *Contributions to Mineralogy and Petrology*. <https://doi.org/10.1007/s00410-025-02209-5>

Zhang, Y., Anderson, N.S.*, Mohr, M.T., Nelson, L.L., Macdonald, F.A., Schmitz, M.D., Thurston, O.G., Guenther, W.R., Karlstrom, K.E., and Swanson-Hysell, N.L., (2024), Paleomagnetism of the southwest Laurentia large igneous province and Cardenas Basalt: pulsed magmatism during rapid late Mesoproterozoic plate motion. *JGR Solid Earth*. <https://doi.org/10.1029/2024JB029036>

Hodgin, E.B., Swanson-Hysell, N.L., Kylander-Clark, A.R.C., Turner, A.C., Stolper, D.A., Ibarra, D.E., Schmitz, M.D., **Zhang, Y.**, Fairchild, L.M., Fuentes, A.J., (2024), One billion years of stability in the North American Midcontinent following two-stage Grenvillian structural inversion. *Tectonics*. <https://doi.org/10.1029/2024TC008415>

Zhang, Y., Hodgin, E.B., Alemu, T., Pierce, J.*, Fuentes, A., and Swanson-Hysell, N.L., (2024), Tracking Rodinia into the Neoproterozoic: new paleomagnetic constraints from the Jacobsville Formation. *Tectonics*. <https://doi.org/10.1029/2023TC007866>

Mohr, M.T., Schmitz, M.D., Swanson-Hysell, N.L., Karlstrom, K.E., Macdonald, F.A., Holland, M.E., **Zhang, Y.**, Anderson, N.*, (2024), High-Precision U-Pb geochronology links magmatism in the SW Laurentia Large Igneous Province and Midcontinent Rift. *Geology*. <https://doi.org/10.1130/G51786.1>

Sapienza, F., Gallo, L.S., **Zhang, Y.**, Vaes, Bram., Domeier, Mathew., Swanson-Hysell, N.L., (2023), Quantitative Analysis of Paleomagnetic Sampling Strategies. *JGR Solid Earth*. <https://doi.org/10.1029/2023jb027211>

Gallo, L. C., Domeier, M., Sapienza, F., Swanson-Hysell, N. L., Vaes, B., **Zhang, Y.**, Arnould, M., Eyster, A., Guñer, D., Király, A., Robert, B., Rolf, T., Shephard, G., and van der Boon, A., (2023), Embracing uncertainty to resolve polar wander: a case study of Cenozoic North America. *Geophysical Research Letters*. <https://doi.org/10.1029/2023GL103436>

Slotznick, S.P., Swanson-Hysell, N.L., **Zhang, Y.**, Clayton, K.E., Wellman, C.H., Tosca, N.J., and Strother, P.K., (2023), Reconstructing the paleoenvironment of an oxygenated Mesoproterozoic shoreline and its record of life. *GSA Bulletin*. <https://doi.org/10.1130/B36634.1>

Pierce, J.*, **Zhang, Y.**, Hodgin, E.B., and Swanson-Hysell, N.L., (2022), Quantifying inclination shallowing and representing flattening uncertainty in sedimentary paleomagnetic poles. *Geochemistry, Geophysics, Geosystems*. <https://doi.org/10.1029/2022GC010682>

Rose, I., **Zhang, Y.**, and Swanson-Hysell, N.L., (2022), Bayesian paleomagnetic Euler pole inversion for paleogeographic reconstruction and analysis. *JGR: Solid Earth*. <https://doi.org/10.1029/2021jb023890>

Zhang, Y., Swanson-Hysell, N.L., Avery, M.S., Fu, R.R., (2022), High geomagnetic field intensity recorded by anorthosite xenoliths requires a strongly powered late Mesoproterozoic geodynamo. *PNAS*. <https://doi.org/10.1073/pnas.2202875119>

Hodgin, E.B., Swanson-Hysell, N.L., DeGraff, J.M., Kylander-Clark, A.R.C., Schmitz, M.D., Turner, A.C., **Zhang, Y.**, Stolper, D.A., (2022), Final inversion of the Midcontinent Rift during the Rigolet Phase of the Grenvillian Orogeny. *Geology*. <https://doi.org/10.1130/G49439.1>

Cromwell, G., **Zhang, Y.**, (2021), New paleointensity data from Aniakchak volcano, Alaska, USA. *Geochemistry, Geophysics, Geosystems*. <https://doi.org/10.1029/2021gc010032>

Zhang, Y., Swanson-Hysell, N.L., Schmitz, M.D., Miller Jr., J.D., and Avery, M.S., (2021), Synchronous emplacement of the anorthosite xenolith-bearing Beaver River diabase and one of the largest lava flows on Earth. *Geochemistry, Geophysics, Geosystems*. <https://doi.org/10.1029/2021GC009909>

Swanson-Hysell, N.L., Avery, M.S., **Zhang, Y.**, Hodgkin, E.B., Sherwood, R.J., Apen, F.E., et al., (2021). The paleogeography of Laurentia in its early years: new constraints from the Paleoproterozoic East Central Minnesota batholith. *Tectonics*. <https://doi.org/10.1029/2021TC0067511>

Swanson-Hysell, N.L., Hoaglund, S.A., Crowley, J.L., Schmitz, M.D., **Zhang, Y.**, and Miller Jr., J.D., (2020), Rapid emplacement of massive Duluth Complex intrusions within the Midcontinent Rift. *Geology*. <https://doi.org/10.1130/g47873.1>

Zhang, Y., Pairing paleointensity results with coercivity spectra: providing support for selection criteria. *IRM Quarterly*. Volume 30. Number 1.

TALKS

2026 GSA Annual Meeting October 11-14, 2026
U-Pb joint inversion for thermal history reconstruction

2026 GSA Annual Meeting October 11-14, 2026
Cryogenian collision and stitching plutonism in the northwestern Arabian-Nubian Shield preceded final closure of the Mozambique Ocean

Space Physics Seminar, University of Minnesota Twin Cities (invited) April 14, 2026
The Geodynamo Through Time: core paradoxes, paleomagnetic insights, and open questions

Geology Seminar, Occidental College (invited) November 4, 2025
Deep time, moving continents, and a snowball Earth

Division Seminar, Caltech (invited) November 3, 2025
Assembling supercontinent Rodinia and tracking its development in the late Proterozoic

2025 GSA Annual Meeting October 19-22, 2025
Recalibrating the history of metamorphism and exhumation in the Adirondack Highlands leads to a major revision of Neoproterozoic global paleogeography

2025 GSA Annual Meeting (invited) October 19-22, 2025
Paleogeographic Evolution of Oman and Egyptian Terranes of the Arabian-Nubian Shield through the Neoproterozoic

Earth & Environmental Sciences Department seminar, UMN Duluth April 8, 2025
Anorthosites in northern Minnesota: 100 years of study of their 1-billion-year-old history

Scripps Institute for Oceanography October 7, 2024
Reconstructing late Proterozoic magmatism, geomagnetic field behavior, and paleogeography by pairing paleomagnetism with state-of-the-art geochronology

Hard Rock Lunch seminar, University of Minnesota Twin Cities October 2, 2024
Reconstructing Neoproterozoic paleogeography with metamorphic rocks

UC Berkeley EPS exit seminar April 25, 2024
Reconstructing late Proterozoic magmatism, geomagnetic field behavior, and paleogeography using tiny magnets in rocks

2023 AGU Fall Meeting Dec 5 2023
Constraining the position of Arabian-Nubian arc terranes in the lead-up to Snowball Earth glaciation: new constraints from Tonian dikes of Oman

Institute of Geology and Geophysics seminar, Chinese Academy of Science (<i>invited</i>) <i>New approaches to APWP synthesis and incorporating uncertainties in sedimentary records</i>	July 5 2023
2023 IRM conference (<i>invited</i>) <i>New approaches to apparent polar wander path development constrain the late Mesoproterozoic assembly of Rodinia</i>	June 5-June 8 2023
2023 MagIC workshop (<i>invited</i>) <i>New perspectives on Laurentia's Grenville Loop: tracking Rodinia across the Mesoproterozoic to Neoproterozoic boundary</i>	Feb 28-Mar 2 2023
2022 AGU Fall Meeting <i>Reconstructing the position of the supercontinent Rodinia in the early Neoproterozoic: new constraints from Laurentia's interior and the Grenville margin</i>	Dec 12-16 2022
Beijing Paleomagnetism and Geochronology Laboratory (<i>invited, online</i>) <i>High geomagnetic field intensity recorded by anorthosite xenoliths requires a strongly powered late Mesoproterozoic geodynamo</i>	Sep 28 2022
Young CEED 21 Frontiers in quantitative paleogeography (<i>invited, online</i>) <i>Bayesian_PEP_inversion: a Bayesian framework for integrating paleomagnetic and geochronologic data into apparent polar wander inversion</i>	Nov 14-20 2021
Grand Canyon Supergroup Field Forum (<i>invited</i>) <i>The rich paleomagnetic record of the Mesoproterozoic Midcontinent Rift and the Southwestern Laurentia Large Igneous Province</i>	April 9-19 2021
Cloud Meeting on Paleomagnetism (<i>online</i>) <i>Intense magmatic activity and a strong geomagnetic field—a study on the anorthosite xenoliths hosted in the Mesoproterozoic Midcontinent Rift diabase</i>	1/29/2021
2020 AGU Conference (<i>online</i>) <i>Recovering Mesoproterozoic geomagnetic field intensity using anorthosite xenoliths hosted in Midcontinent Rift diabase</i>	12/15/2020
North Central GSA Conference, Duluth, MN (<i>online</i>) <i>The rich paleomagnetic records of Proterozoic Midcontinent Rift intrusives: an updated synthesis with a new pole from the Beaver River diabase</i>	05/18/2019
Institute for Rock Magnetism, University of Minnesota, Minneapolis, MN (<i>invited</i>) <i>Paleomagnetism and rock magnetism on the Beaver River diabase and anorthosite xenoliths therein</i>	01/09/2019

TEACHING

Lecturer at the IRM summer school <i>micromagnetic modeling & magnetic microscopy</i> https://duserzym.github.io/rock_magnetometry/	July 20226
Teaching assistant for lab sessions at the IRM summer school	June 2024
Graduate student instructor (GSI) for EPS 101 Field Geology and Digital Mapping <i>Advisor: Matthew Gleeson</i>	Fall 2023

Reader for EPS 88 PyEarth: A Python Introduction to Earth Science <i>Advisor: Nicholas Swanson-Hysell</i>	Spring 2023
GSI for EPS 101 Field Geology and Digital Mapping <i>Advisor: Nicholas Swanson-Hysell</i>	Fall 2022
Reader for EPS 115 Stratigraphy and Earth History <i>Advisor: Eben Blake Hodgins</i>	Spring 2022
GSI for EPS 101 Field Geology and Digital Mapping <i>Advisor: Nicholas Swanson-Hysell</i>	Fall 2021
GSI for EPS 50 The Planet Earth <i>Advisor: Daniel Stolper</i>	Spring 2021
Participant in the Graduate Remote Instruction Innovation Fellows Program	Winter 2020
GSI for EPS 101 Field Geology and Digital Mapping <i>Advisor: Nicholas Swanson-Hysell</i>	Fall 2020
Completion of UC Berkeley GSI Conference training	Jan 2020
Completion of required Online Course: Professional Standards and Ethics for GSIs	Fall 2019
Completion of required Pedagogy Course EPS 375	Fall 2019

ORIGINAL GEOLOGICAL FIELD WORK

Southern Sweden, [1 week] <i>Reconstructing the paleogeographic position of Baltica in the Neoproterozoic</i>	2026
Central Eastern Desert, Egypt [2 weeks] <i>Reconstructing the Neoproterozoic paleogeography of the Arabian-Nubian shield</i>	2025
Sliderock Mountains, Montana [1 week] <i>Extending site-based apparent polar wander path for North America into the Mesozoic with paleomagnetic data from the intrusive rocks of the Sliderock Mountain Volcanics</i>	2024
Oman [5 weeks] <i>Constraining the paleogeographic configuration of the Mozambique Ocean in the Neoproterozoic</i>	2024, 2026
Grenville orogen [3 weeks] <i>Pairing paleomagnetic data and thermochronology records to reevaluate the exhumation history of rocks of the Grenvillian Orogeny and recalibrate the Grenville Loop</i>	2022, 2023
Death Valley, California; Grand Canyon, Arizona [4 weeks] <i>Using paleomagnetism and geochronology to study the temporal and magmatic relationship between the ca. 1.1 Ga South-western Laurentia Large Igneous Province and the Midcontinent Rift</i>	2021
Pikes Peak, Colorado [2 weeks] <i>Using paleomagnetism and geochronology to study the emplacement history of Pikes Peak batholith and its temporal and magmatic associations with the Midcontinent Rift 1.1 billion years ago</i>	2020

Midcontinent Rift, Lake Superior Region [11 weeks] 2019, 2020, 2021
Reconstructing the behavior of the Mesoproterozoic geomagnetic field, continental motion during supercontinent assembly and the nature of ancient environments through paleomagnetic studies of the intrusive rocks and sediments of the 1.1 billion-year-old Midcontinent Rift

Central Highland, Iceland [3 weeks] 2019
Qualitatively and quantitatively measure the erosion rate of rofabards soil erosion in Central highlands, Iceland

AWARDS AND FUNDING

IAGA Early Career Presentation Awards 2025

AGU GPE Post-doc grant (\$755) 2024
Enhancing site-based apparent polar wander paths with Bayesian stratigraphic age models

UC Berkeley graduate student conference travel grant (\$900) 2023
2023 Geochronology Gordon Research Conference

UC Berkeley Earth and Planetary Science George C. Louderback Award 2023

2023 AGES3-Grad Geochronology Award (\$8,865) 2023
Dating the Grenville Loop using U-Pb apatite thermochronology

2022 GSA Graduate Student Research Grant (\$1,749) 2022
Paleomagnetism and thermochronology of the Adirondack Mountains, Grenville Province

UC Berkeley graduate student conference travel grant (\$900) 2022
2022 AGU Fall Meeting oral presentation

Hearts to Humanity Eternal (H2H8) Programs (\$10,000) 2022
H2H8 Association Graduate Research Grant to Advance Humanity through Science

U.S. Visiting Student Fellowships, Institute for Rock Magnetism 2019
Paleomagnetism and rock magnetism study on Mesoproterozoic Beaver Bay Complex and anorthosite xenoliths therein (\$500)

ILSG Student Research Fund, Institute on Lake Superior Geology 2019
To study the emplacement history of the Beaver River diabase and the anorthosite xenoliths therein using paleomagnetism (\$500)

Chevron-Xenel Gateway Fellowship, Berkeley International House (\$5,000) 2019

John Parke Young Student Grant, Occidental College 2019
Using UAV Data for Monitoring Modern Rofabard Soil Erosion in Central Highlands, Iceland (\$3,500)

Independent research, Occidental College 2018
Pseudo-Thellier Paleointensity Measurement on R-N Geomagnetic Polarity Reversal Recorded by Mafic Lava Flows, Anahola, Kauai (\$4,000)

Independent research, Henry Luce Foundation, Nanjing University 2017
Mapping of Ambient Ozone Pollution in China and the Assessment of Its Health Impact on Socio-Economy (\$2,250)

SERVICE

Contributing developer to the open source PmagPy software project (<https://github.com/PmagPy/PmagPy>).

Contributing developer to the open source RockmagPy software project (<https://github.com/PmagPy/RockmagPy-notebooks>).

Lead developer for the RapidPy superconducting rock magnetometer automatic sample changing system control software (<https://duserzym.github.io/RAPID/>).

IN THE NEWS

PNAS In This Issue: Evidence of strong Proterozoic magnetic field
<https://www.pnas.org/doi/10.1073/iti2922119>

EOS news: Swinging Strength of Earth's Magnetic Field Could Signal Inner Core Formation
<https://eos.org/articles/swinging-strength-of-earths-magnetic-field-could-signal-inner-core-formation>

UC Berkeley EPS department news: EPS Research on Earth's Ancient Magnetic Field Featured in EOS:
<https://eps.berkeley.edu/news/eps-research-earths-ancient-magnetic-field-featured-eos>

Diamonds Are a Paleomagnetist's Best Friend in EOS:
<https://eos.org/articles/diamonds-are-a-paleomagnetists-best-friend>

Reviewer for the following journals

Earth and Planetary Science Letters
Geochemistry, Geophysics, Geosystems
GEOLOGY
Nature Communications Earth & Environment
Science Bulletin
Tectonics
Science Advances

Reviewer for the following panels

NASA Emerging Worlds
National Science Foundation

MEMBERSHIPS

American Geophysical Union (AGU)
Geological Society of America (GSA)